VBF-T2R1NOFORCE-DVPR-001 - Design Verification Report (F1)

Design Verification Plan and Report (Virtual Test Version) - F1

VBF-T2R1NOFORCE-DVPR-001 | Case: t2_r1_noforce | Date: 2026-08-25 | All results from simulation outputs; uncovered conditions honestly marked N/A.

#	Verification item	Condition	Result	Determination	Source
1	1C discharge capacity	1C CC discharge, 25 °C	5.0282 Ah (nominal 5.0 Ah)	PASS (>= nominal; no explicit capacity criterion in task)	`cell/r5_f1_1c_spme.json:capacity_ah`
2	Energy density	contract caliber (ED = ?V·I dt / ? layer mass, electrolyte excluded)	435.44 Wh/kg	**PASS (>= 327.18 Wh/kg)**	`cell/r5_f1_energy.json:energy_density_wh_kg`
3	4C fast-charge plating	4C charge from 45 °C, anode potential criterion	min anode potential +0.0071 V	**PASS (no plating, phi >= 0 V)**	`cell/r5_f1_4c_dfn.json:anode_potential_v` (min, tool-derived)
4	4C fast-charge temperature rise	4C charge, lumped thermal, h=10	T_max 359.92 K (+41.8 K from 45 °C)	reported (no temperature threshold in task criteria)	`cell/r5_f1_4c_dfn.json:T_max_K`
5	-20 °C discharge retention	1C discharge, 253.15 K ambient + initial temp (cold-start fix)	99.45 % (5.0003 / 5.0282 Ah)	**PASS (>= 90 %)**	`derived/r5_f1_retention.json` ? `cell/r5_f1_lowT_spme.json`
6	SEI thickness @100 cyc	1C cycling, 100 cyc, 25 °C isothermal	439.52 nm	**PASS (<= 500 nm)**	`cell/r5_f1_aging100_spme.json:sei_thickness_nm_end`
7	SEI thickness @500 cyc	1C cycling, 500 cyc, 25 °C isothermal	738.43 nm	**FAIL (> 550 nm)** - documented unreachable (three-strike, final entry)	`cell/r5_f1_aging500_spme.json:sei_thickness_nm_end`
8	Voltage window	parameter set limits	2.5 - 4.2 V	PASS (usage within limits)	parameter set cut-offs

Items explicitly N/A (beyond pure-simulation boundary, require physical experiment): nail penetration, overcharge-to-thermal-runaway, crush, drop, rate-pulse internal resistance (DCR is model-computed only), cycle-life capacity retention (aging simulated for SEI growth only; capacity trajectory shows the climb-then-saturate artifact and is not reported as degradation), electrode manufacturability tolerances.

Conclusion: 4 of 5 task criteria PASS (ED, SEI@100, -20 °C retention, 4C no-plating). 1 criterion FAIL (SEI@500 <= 550 nm) - not a design failure but a model-family limitation: the ec-reaction-limited SEI growth law cannot saturate; measured minimum growth ratio 100->500 cyc is x1.68 and the k_SEI lever saturates (R5 F2). Uncovered items listed above are cited as paper limitations.